Towards Versatile Opti-Acoustic Sensor Fusion and Volumetric Mapping for Safe Underwater Navigation

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Motivation

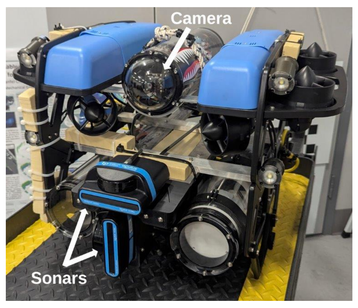
Tasks such as maintenance demand **accurate close-range** sensing to guide manipulators and avoid unmodeled obstacles.
Reliable sensing and mapping across **diverse conditions** is critical for AUV autonomy and safety.

PROBLEM

➔ **Monocular vision**-based reconstruction methods are **unreliable in turbid waters** and **lack depth scale** information.
➔ **Sonars** are robust to turbid water and non-uniform lighting conditions, however, they have **low resolution** and **elevation ambiguity**.

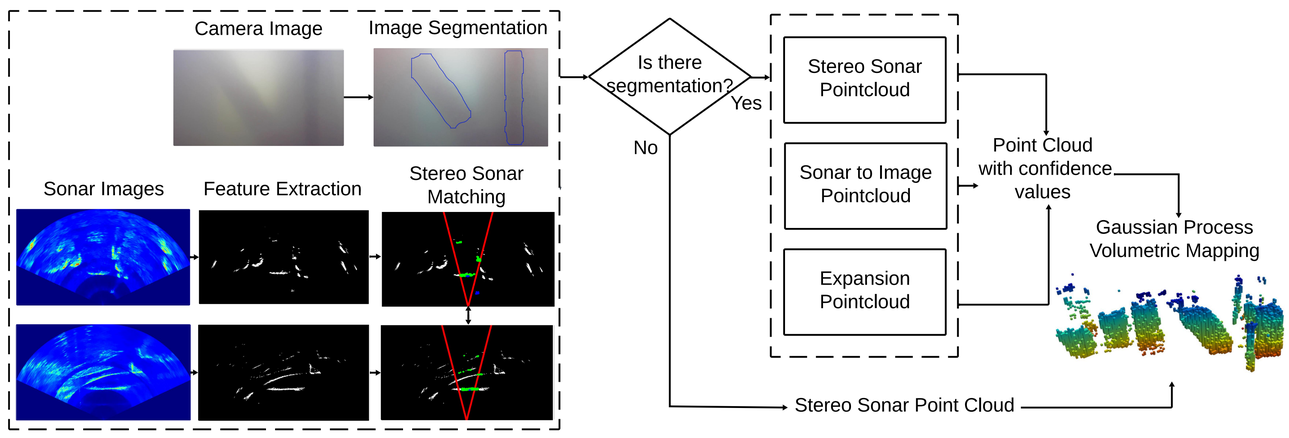
GOAL

Provide a **sensor fusion** and **mapping** method that:
➔ Applicable in **diverse settings** (turbidity).
➔ Suitable for platforms running in **real-time.**
➔ Does not require **sonar datasets** or **multi-view** data.
➔ Prioritizes **reliable** and **risk-sensitive** (close-range) data.
➔ Reconstructs **complex geometries**.

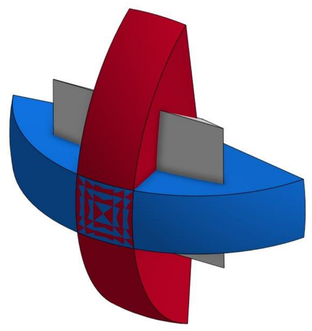


Methodology

GENERAL PIPELINE



We assume that a robot is equipped with two orthogonal **forward-looking imaging sonars** and a **forward-looking camera**, with **overlapping fields of view**

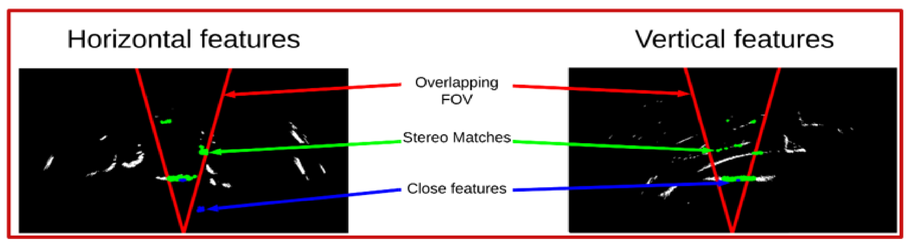


SONAR TO IMAGE MATCHING

Leveraging their overlapping vantage points, each lacking a dimension in their respective coordinate frames.
➔ **Distance** is derived from the original **sonar return**
➔ **Elevation** is determined by the **optical image**

STEREO SONAR MATCHING

Limited FOV bounded by vertical beam width.



CONFIDENCE VALUES

Confidence values are assigned to resulting 3D points based on **fusion method reliability**.

GAUSSIAN PROCESS MAPPING

Continuous distribution model, can extrapolate across sparse data.
Heteroscedastic GP, assigning each observation its own noise variance value that reflects the reliability of its source.
Observation **noise values** are inversely proportional to **confidence values**.

Computation Time Results

Mapping time in (seconds) with standard deviation. Environment example images.



Method	Tank Single Disk	Tank Double Disk	Marina Pier
Octo SS RGB	0.1495 ± 0.0838	0.1168 ± 0.0414	0.2797 ± 0.3474
GP SS RGB	**0.0551** ± 0.0634	**0.0699** ± 0.0363	0.1040 ± 0.0365
GPC SS RGB	0.0555 ± **0.0197**	0.0712 ± **0.0311**	**0.0970** ± **0.0324**

Conclusion

Performance against state-of-the-art techniques across scenes with **varying geometric complexity**, demonstrates **superior accuracy** while maintaining adequate **scene coverage**. Field experiments further demonstrate the system's ability to map both small and large structures in **highly turbid** conditions.

This sensor fusion prioritizes mapping **close-range**, and **critical features** for **safe navigation** and manipulation under varying visibility .

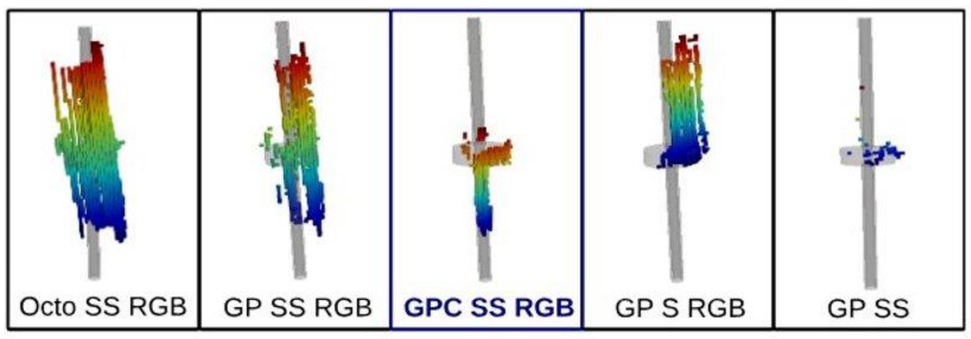
Mapping Results

Voxel map results.

Voxel colors depict height.

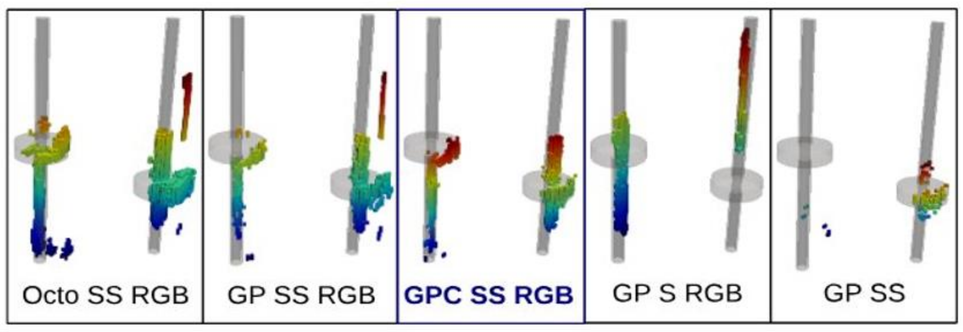
Proposed approach highlighted **blue** in image and **gray** in tables.

TANK SINGLE DISK RESULTS



Method	MAE (cm) ↓	RMSE (cm) ↓	SD (cm) ↓	Precision (%) ↑	Inlier Voxels ↑
Octo SS RGB	5.9844	0.8106	0.5468	34.5989	**1212**
GP SS RGB	6.6030	0.8816	0.5841	35.7711	450
GPC SS RGB	**1.1712**	**0.1579**	**0.1059**	**88.1844**	306
GP S RGB	6.0007	0.8305	0.5741	38.8336	273
GP SS	1.3590	0.1819	0.1209	81.5789	31

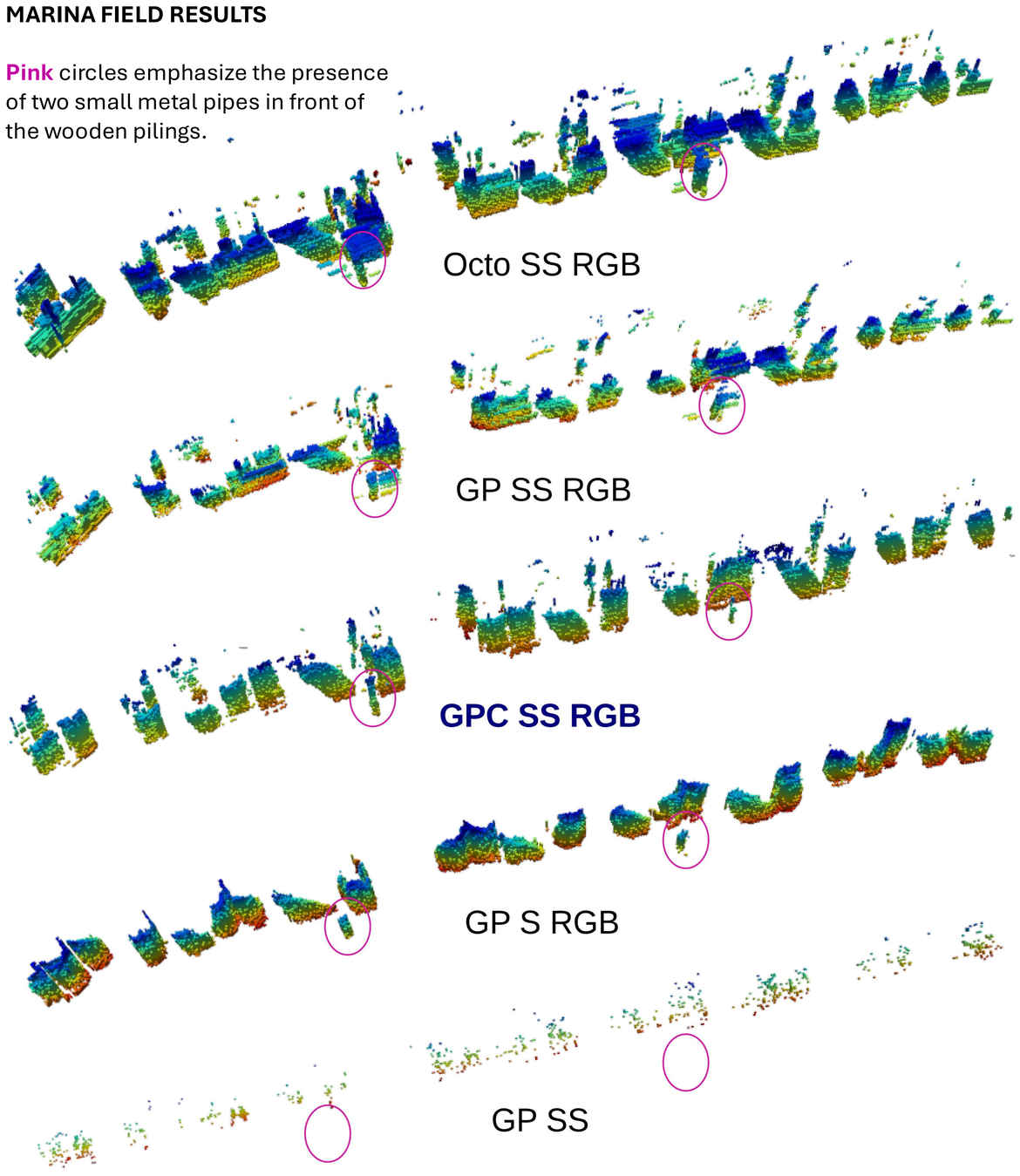
TANK DOUBLE DISK RESULTS



Method	MAE (cm) ↓	SD (cm) ↓	RMSE (cm) ↓	Precision (%) ↑	Inlier Voxels ↑
Octo SS RGB	2.6064	0.3141	0.4081	64.1379	**1302**
GP SS RGB	2.5962	0.3338	0.4229	68.8243	521
GPC SS RGB	1.6219	0.1651	0.2315	**81.5825**	598
GP S RGB	**1.3885**	**0.1306**	**0.1906**	79.5385	517
GP SS	2.3853	0.3511	0.4244	68.4685	76

MARINA FIELD RESULTS

Pink circles emphasize the presence of two small metal pipes in front of the wooden pilings.



Acknowledgements
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